風溼免疫科常見檢查及檢驗介紹

(蔡明翰整理)

Antinuclear Antibodies (ANA) (抗核抗體):

Negative: A useful screening test in patients with symptoms suggesting collagen-vascular disease, especially if titer is >=1:160.
Positive: Systemic lupus erythematosus (SLE), scleroderma, mixed connective tissue disease (MCTD), RA, polymyositis etc. Low titers are seen in non-collagen-vascular disease.

ANA patterns:

(1) Homogeneous (均質型): Nonspecific. Seen in SLE and a variety of diseases.

(2) Speckled (點狀型): Seen in many connective tissue disorders.

(3) Peripheral (週邊型): Specific in SLE.(less)

(4) Nucleolar (核仁型): Positive in Sjogren's syndrome and scleroderma.(less)

Anti-Neutrophil Cytoplasmic Antibodies (ANCAs) (抗中性球細胞漿抗體):

Seen in vasculitis (ex: Wegener's granulomatosis) (韋氏肉芽腫) (自費 NT 1000)

Antiphospholipid Antibodies (抗磷脂質抗體):

They are associated with recurrent arterial or venous thrombosis, young stroke, or spontaneous abortion. There are two antibodies:

1. Anticardiolipin antibody (抗心脂質抗體):

(1) IgG: (-):<12 GPL unit/ ml;
(+-): 12-18
(+): >18

(2) IgM: 立即短暫出現之免疫球蛋白，判讀單位同 IgG.

2. Lupus Anticoagulant (LA) (狼瘡抗凝血時間):

Normal: 31-44 sec. (>44 sec: may have antiphospholipid syndrome)

Anti-double stranded DNA antibodies (Anti-ds-DNA Ab) (抗雙股 DNA 抗體):

They are associated with SLE disease activity and lupus nephritis. If the activity is high, they should be checked every month to follow up the effect of treatment.

Anti-ds DNA 的值和 SLE 疾病活動度相關；而 ANA 則否。

Anti-Extractable Nuclear Antibodies (Anti-ENA Ab) (抗可萃取核抗體):

If ANA is positive and the patient is suspected as SLE, we may order them.
They includes:
(1) Anti- Sm Ab: specific to SLE
(2) Anti-RNP Ab: MCTD or SLE
(3) Anti- Ro Ab (SS-A): SLE or Sjogren's syndrome
(4) Anti-La Ab (SS-B): SLE or Sjogren's syndrome
(5) Anti-Scl. 70: SLE or sclerosis

Complement (C3, C4) (補體):

和免疫有關的蛋白質，SLE 活動度高時會下降。

(1) C3: Normal: 73-134 mg/dl
(2) C4: Normal: 18.2- 45.5 mg/dl

Cortisol (可體素):

懷疑病人服用過量類固醇 (如: 黑藥丸等)有滿月臉、水牛肩等症狀時測。

Normal (serum): 8am: 5.0-23.0 ug/dl; 4pm: 3.0-15.0 ug/dl

Creatine Phosphokinase (CPK) (肌酸磷酸激酵素):

It may increase in AMI, rhabdomyolysis, and polymyositis (多發性肌炎) etc.

Normal: 25-150 U/L

C-Reactive Protein (CRP) (C- 反應蛋白): (急性發炎指數)

Increased: In infections (especially in bacterial infection), tissue injury or necrosis, and inflammatory disorders (ex: SLE, RA, vasculitis etc.)
Normal: <5 mg/L

Cryoglobulins (冷凝球蛋白):

Immune complex syndrome ex: SLE
Normal: negative

Erythrocyte Sedimentation Rate (ESR) (紅血球沉降速率): (發炎指數)

Increased: Active RA, SLE etc.
Normal: male: 0-15 mm/hr; female: 0-20 mm/hr

Human Leukocyte Antigen (HLA) (人類白血球抗原):

在第六對染色體上，許多自體免疫疾病與之有關 如: 僵直性脊椎炎常有 HLA B27 (+); RA: HLA DR4 etc.

Protein C:
Associated with vascular thrombosis or CVA (level decreased)
Normal: 73-107%

Protein S:
Associated with vascular thrombosis or CVA (level decreased)
Normal: 90-130%

Rheumatoid Factor (RF) (類風溼因子):

RF is found in the serum of 75% to 80% of RA patients. It may be present in SLE or endocarditis. RF is not necessary to diagnose RA.
Normal: <40 mg/dl

Urid Acid (尿酸):

Gout acute attack 不一定尿酸高，反之亦然，需靠臨床症狀判斷。

Normal: 4.5-8.2 mg/dl

VDRL and RPR:

Some SLE patients may have false- positive serologic tests for syphilis.

Normal: negative